

A few more

Diffusion articles

DDIM

[cs.LG] 5 Oct 2022

DENOISING DIFFUSION IMPLICIT MODELS

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ABSTRACT

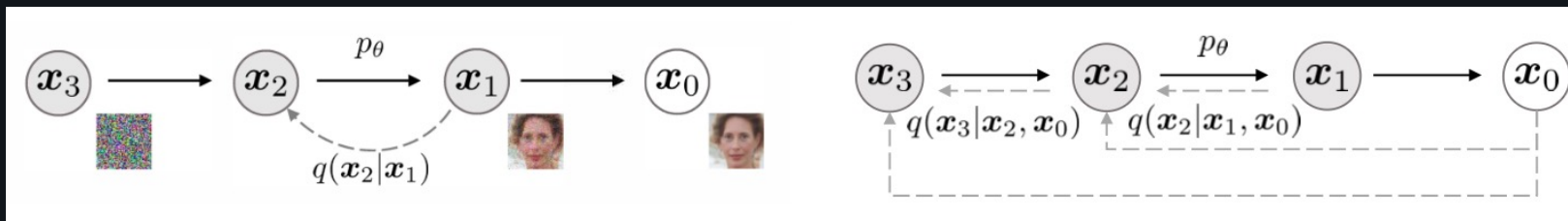
Denoising diffusion probabilistic models (DDPMs) have achieved high quality image generation without adversarial training, yet they require simulating a Markov chain for many steps in order to produce a sample. To accelerate sampling, we present denoising diffusion implicit models (DDIMs), a more efficient class of iterative implicit probabilistic models with the same training procedure as DDPMs. In DDPMs, the generative process is defined as the reverse of a particular Markovian diffusion process. We generalize DDPMs via a class of non-Markovian diffusion processes that lead to the same training objective. These non-Markovian processes can correspond to generative processes that are deterministic, giving rise to implicit models that produce high quality samples much faster. We empirically demonstrate that DDIMs can produce high quality samples $10\times$ to $50\times$ faster in terms of wall-clock time compared to DDPMs, allow us to trade off computation for sample quality, perform semantically meaningful image interpolation directly in the latent space, and reconstruct observations with very low error.

DDIM – были

DDPM, Score-Based – много итераций на семпл

GAN – трудно учить, плохо покрывает моды

DDIM – суть



DDPM ~1000 итераций

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}),$$



$$q_\sigma(\mathbf{x}_{1:T}|\mathbf{x}_0) := q_\sigma(\mathbf{x}_T|\mathbf{x}_0) \prod_{t=2}^T q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$$

where $q(\mathbf{x}_t|\mathbf{x}_{t-1}) := \mathcal{N}\left(\sqrt{\frac{\alpha_t}{\alpha_{t-1}}}\mathbf{x}_{t-1}, \left(1 - \frac{\alpha_t}{\alpha_{t-1}}\right)\mathbf{I}\right)$

where $q_\sigma(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\mathbf{x}_0, (1 - \alpha_T)\mathbf{I})$ and for all $t > 1$,

$$q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}}\mathbf{x}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2\mathbf{I}\right).$$

$$L_\gamma(\epsilon_\theta) := \sum_{t=1}^T \gamma_t \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \left[\|\epsilon_\theta^{(t)}(\sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1 - \alpha_t}\epsilon_t) - \epsilon_t\|_2^2 \right]$$

$$q_\sigma(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_0, (1 - \alpha_t)\mathbf{I})$$

- Не марковский процесс
- Маргиналы те же
- Loss тот же

DDIM

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon$$



$\hat{\mathbf{x}}_0 =$

$$f_{\theta}^{(t)}(\mathbf{x}_t) := (\mathbf{x}_t - \sqrt{1 - \alpha_t} \cdot \epsilon_{\theta}^{(t)}(\mathbf{x}_t)) / \sqrt{\alpha_t}.$$

Семплинг

$$p_{\theta}^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \begin{cases} \mathcal{N}(f_{\theta}^{(1)}(\mathbf{x}_1), \sigma_1^2 \mathbf{I}) & \text{if } t = 1 \\ q_{\sigma}(\mathbf{x}_{t-1} | \mathbf{x}_t, f_{\theta}^{(t)}(\mathbf{x}_t)) & \text{otherwise,} \end{cases}$$

Гаусовская (вычисляется)

$$\mathbf{x}_{t-1} = \underbrace{\sqrt{\alpha_{t-1}} \left(\frac{\mathbf{x}_t - \sqrt{1 - \alpha_t} \epsilon_{\theta}^{(t)}(\mathbf{x}_t)}{\sqrt{\alpha_t}} \right)}_{\text{"predicted } \mathbf{x}_0"} + \underbrace{\sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_{\theta}^{(t)}(\mathbf{x}_t)}_{\text{"direction pointing to } \mathbf{x}_t"} + \underbrace{\sigma_t \epsilon_t}_{\text{random noise}}$$

Оптимизация эквивалентна оптимизации DDPM

$$J_{\sigma}(\epsilon_{\theta}) := \mathbb{E}_{\mathbf{x}_{0:T} \sim q_{\sigma}(\mathbf{x}_{0:T})} [\log q_{\sigma}(\mathbf{x}_{1:T} | \mathbf{x}_0) - \log p_{\theta}(\mathbf{x}_{0:T})]$$

$$J_{\sigma} = L_{\gamma} + C.$$



Можем переиспользовать обученные DDPM для таких не марковских процессов разшумления

DDIM - прикольности

Если

$$\sigma_t = \sqrt{\frac{1 - \alpha_{t-1}}{1 - \alpha_t}} \sqrt{1 - \frac{\alpha_t}{\alpha_{t-1}}}$$

То получаем в точности DDPM

Если

$$\sigma_t = 0$$

То прямой и обратный процессы становятся детерминированными. Это и есть DDIM

Получили частный случай Variance Exploding

$$d\bar{\mathbf{x}}(t) = \epsilon_{\theta}^{(t)} \left(\frac{\bar{\mathbf{x}}(t)}{\sqrt{\sigma^2 + 1}} \right) d\sigma(t)$$

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} d\mathbf{w}.$$

Но по другому ”параметру” решили диффур и семплирование другое, меньше зависит от расстояния между α_t и $\alpha_{t-\Delta t}$
В итоге можно делать меньше шагов

DDIM – speed up 10 - 100x

$\mathbf{x}_{1:T}$



$\{\mathbf{x}_{\tau_1}, \dots, \mathbf{x}_{\tau_S}\}$

$$q(\mathbf{x}_{\tau_i} | \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_{\tau_i}}\mathbf{x}_0, (1 - \alpha_{\tau_i})I)$$

Для примера рассматривают σ интерполирующие от DDIM к DDPM

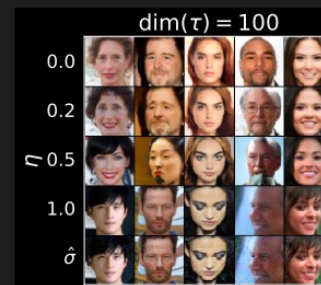
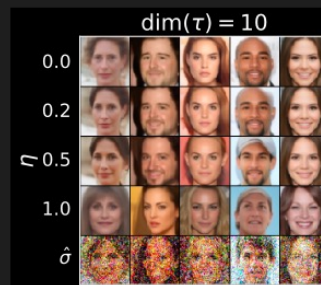
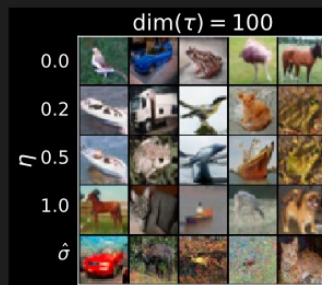
$$\sigma_{\tau_i}(\eta) = \eta \sqrt{\frac{1 - \alpha_{\tau_{i-1}}}{1 - \alpha_{\tau_i}}} \sqrt{1 - \frac{\alpha_{\tau_i}}{\alpha_{\tau_{i-1}}}},$$

$$\hat{\sigma}_{\tau_i} = \sqrt{1 - \alpha_{\tau_i} / \alpha_{\tau_{i-1}}} \quad \text{Скип шагов из Improved DDPM}$$

DDIM

DDPM

S	CIFAR10 (32×32)					CelebA (64×64)					
	10	20	50	100	1000	10	20	50	100	1000	
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
$\hat{\sigma}$	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26	



DDIM

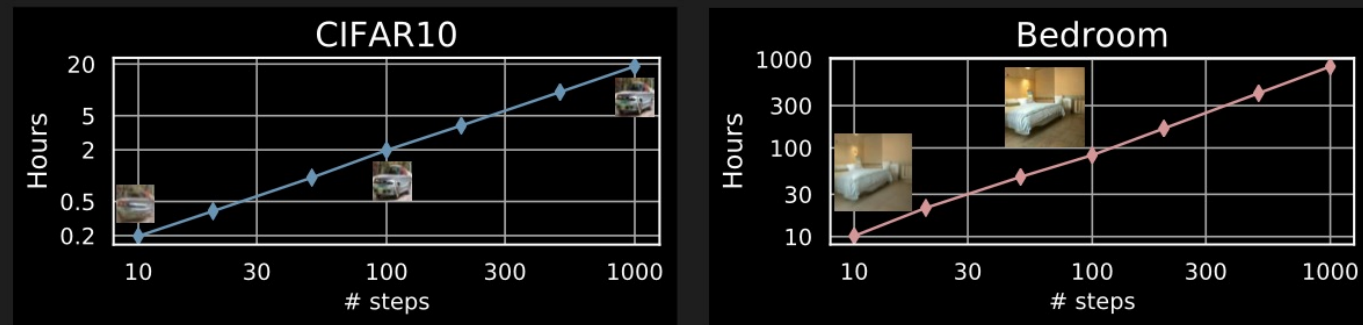
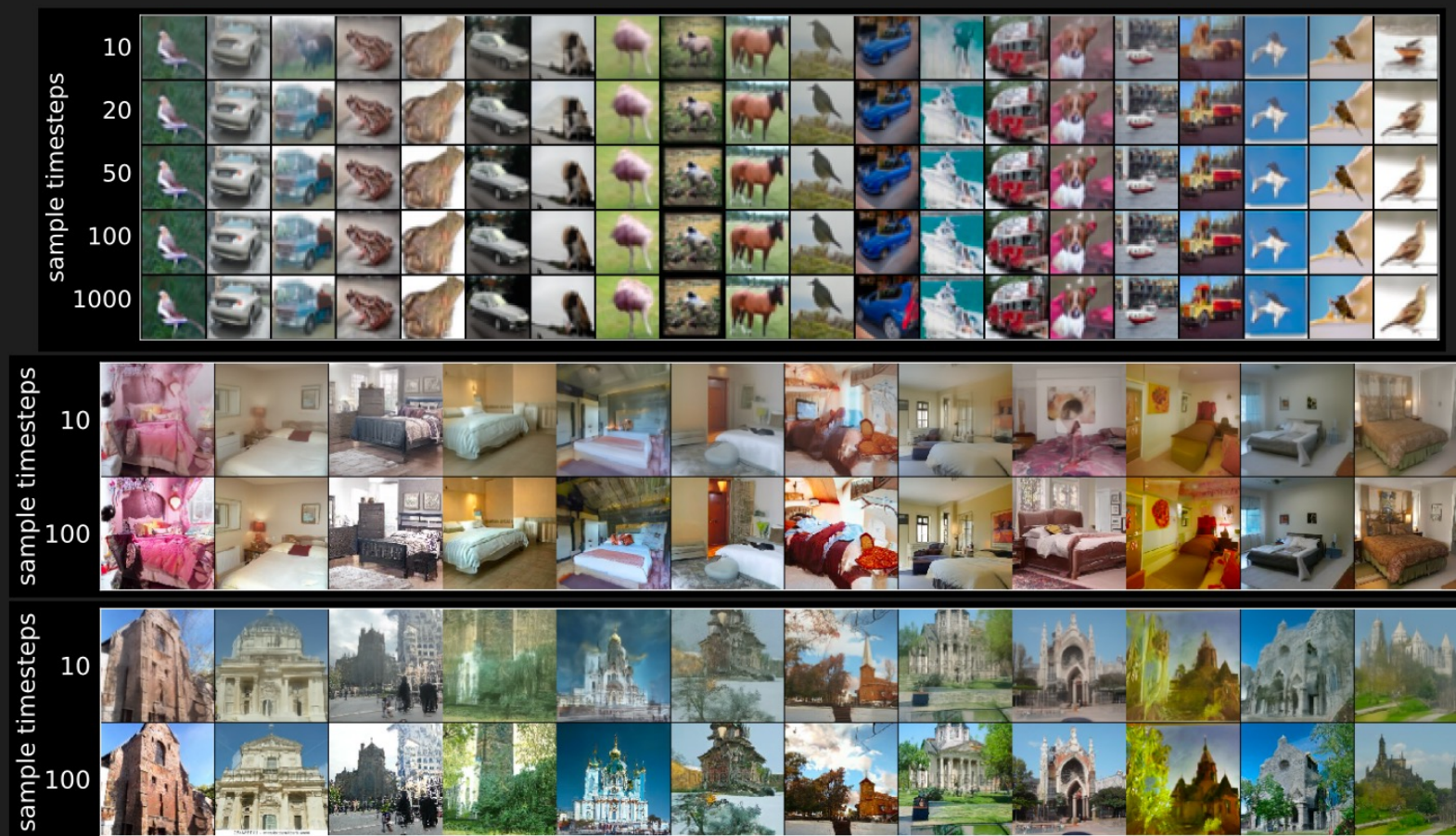


Figure 4: Hours to sample 50k images with one Nvidia 2080 Ti GPU and samples at different steps.

DDIM



Кажется x_t кодирует основное содержание изображение кроме может мелких деталей

DDIM

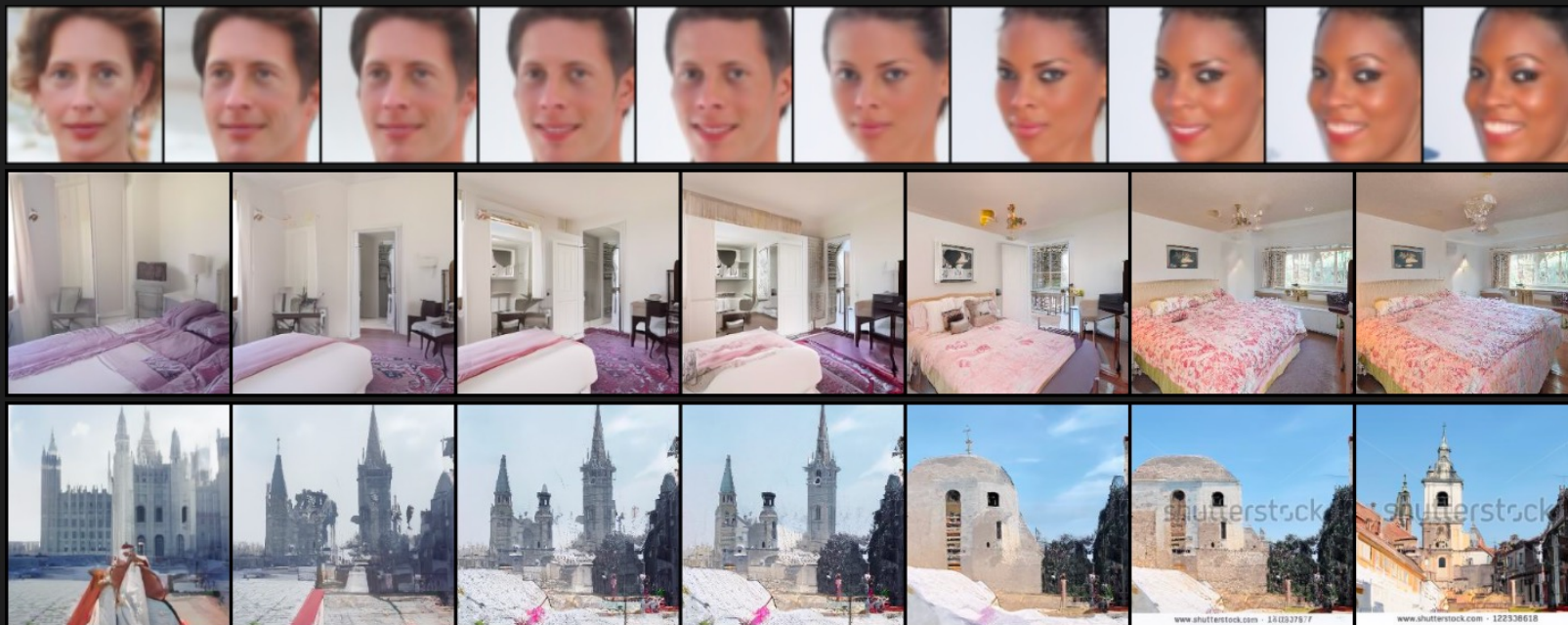


Figure 6: Interpolation of samples from DDIM with $\dim(\tau) = 50$.

В GANах такое есть, ухты, не знал)

DDIM

Table 2: Reconstruction error with DDIM on CIFAR-10 test set, rounded to 10^{-4} .

S	10	20	50	100	200	500	1000
Error	0.014	0.0065	0.0023	0.0009	0.0004	0.0001	0.0001

DDIM можно использовать как кодировщик картинок

DDIM – итоги

- Ввели метод использования обучных DDPM подобных моделей
- Связали его с ODE
- Ввели DDIM – детерминированный в обе стороны
- Показали, что метод можно интерполировать от DDIM к DDPM
- 10 – 100x
- Показали, что латент кодирует верхнеуровневые признаки
- Поинтерполировали латенты
- Показали что латенты довольно круто кодируют картинку
- И это все на 2080 TI



UnifyEdit

Tuning-Free Image Editing with Fidelity and Editability via Unified Latent Diffusion Model

Qi Mao, Lan Chen, Yuchao Gu, Mike Zheng Shou, Ming-Hsuan Yang

Abstract—Balancing fidelity and editability is essential in text-based image editing (TIE), where failures commonly lead to over- or under-editing issues. Existing methods typically rely on attention injections for structure preservation and leverage the inherent text alignment capabilities of pre-trained text-to-image (T2I) models for editability, but they lack explicit and unified mechanisms to properly balance these two objectives. In this work, we introduce *UnifyEdit*, a tuning-free method that performs *diffusion latent optimization* to enable a balanced integration of fidelity and editability within a *unified* framework. Unlike direct attention injections, we develop two attention-based constraints: a self-attention (SA) preservation constraint for structural fidelity, and a cross-attention (CA) alignment constraint to enhance text alignment for improved editability. However, simultaneously applying both constraints can lead to gradient conflicts, where the dominance of one constraint results in over- or under-editing. To address this challenge, we introduce an adaptive time-step scheduler that dynamically adjusts the influence of these constraints, guiding the diffusion latent toward an optimal balance. Extensive quantitative and qualitative experiments validate the effectiveness of our approach, demonstrating its superiority in achieving a robust balance between structure preservation and text alignment across various editing tasks, outperforming other state-of-the-art methods. The source code will be available at <https://github.com/CUC-MIPG/UnifyEdit>.

Index Terms—Text-based image editing, diffusion model, latent optimization, attention-based constraint, tuning-free.

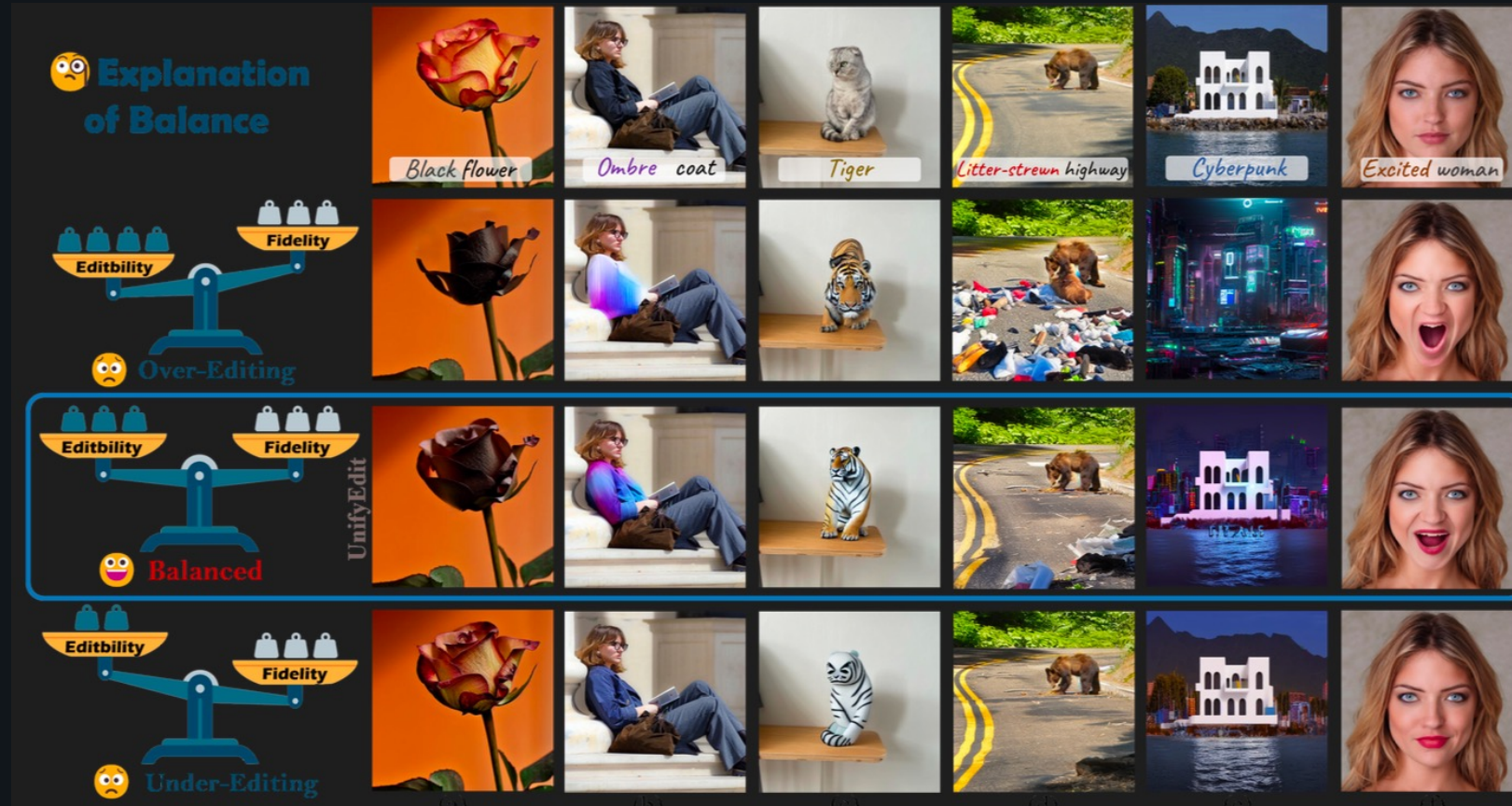


cs.CV] 8 Apr 2025

1 INTRODUCTION

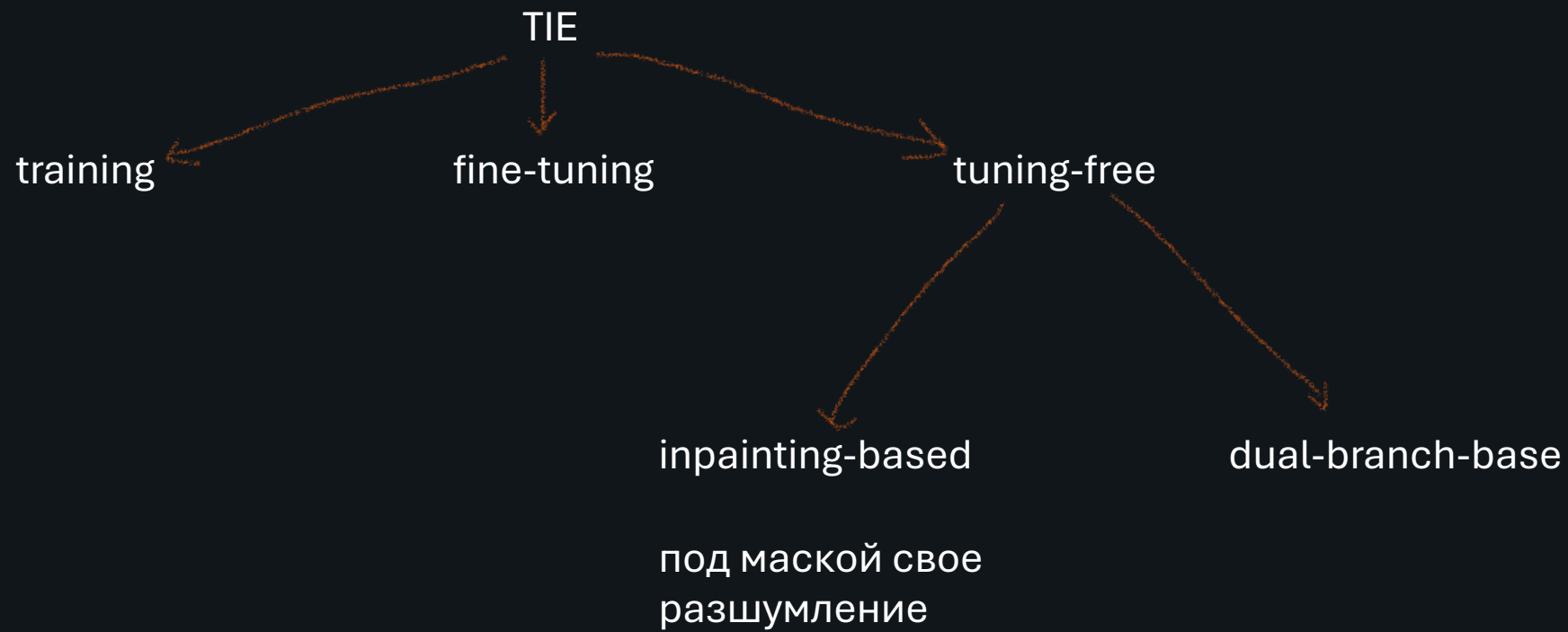
that generates the target image guided by the target prompt. Within

UnifyEdit



Fidelity / Editability

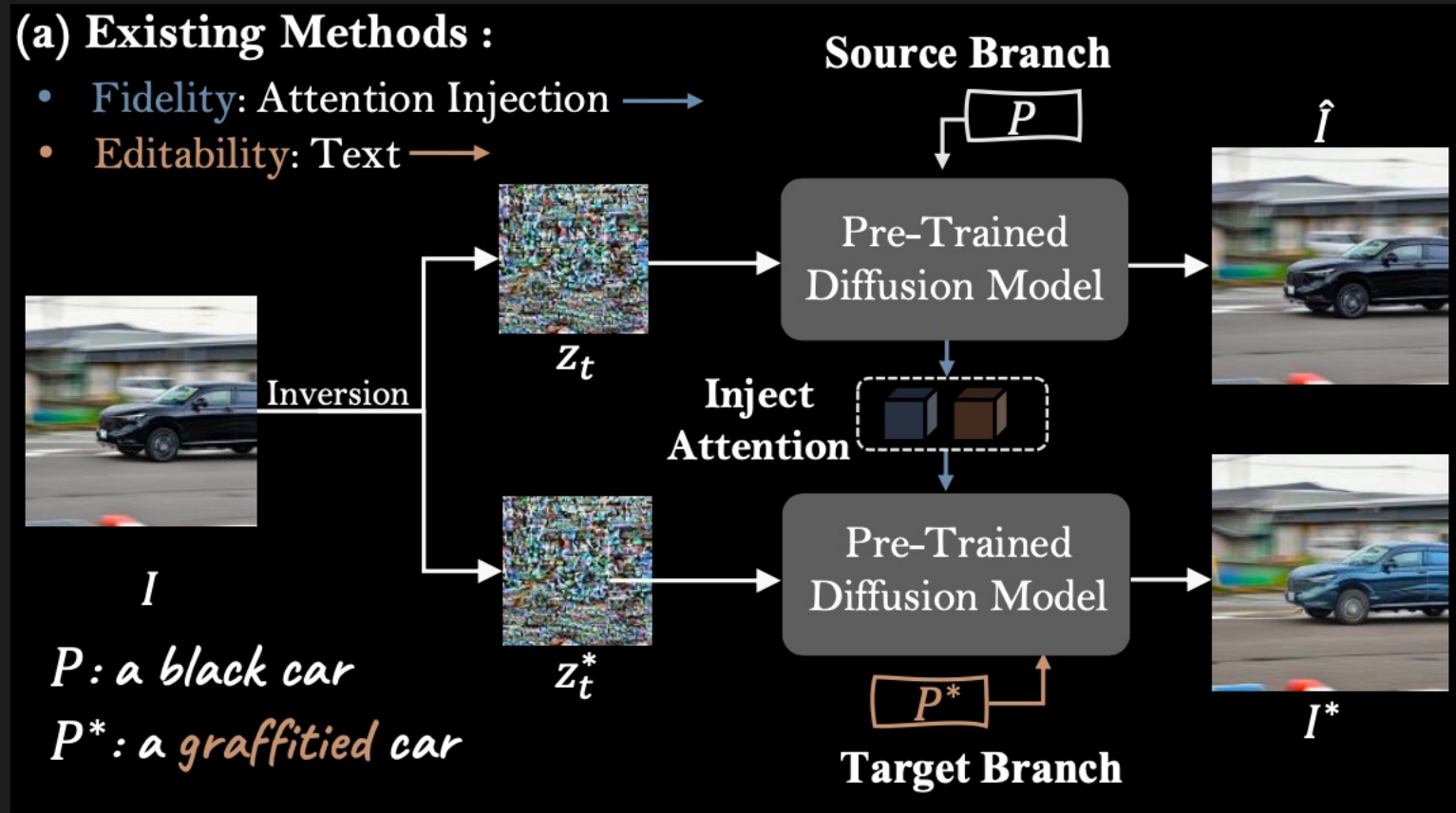
UnifyEdit



UnifyEdit – dual branch

(a) Existing Methods :

- Fidelity: Attention Injection $\rightarrow$
- Editability: Text $\rightarrow$



$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,$$

$$\begin{aligned} A^{*self} &\leftarrow A^{self}, \\ A^{*cross} &\leftarrow A^{cross}. \end{aligned}$$

mainly regulate balance by adjusting attention injection timestep hyperparameters

UnifyEdit

$$\hat{z}_t^* = z_t^* - \mathcal{M} \odot \nabla_{z_t^*} \mathcal{L},$$

$$A^{*self} \leftarrow A^{self}$$



$$\mathcal{L}_{\text{SAP}} = \sum (A_t^{self} - A_t^{*self})^2.$$



$$\mathcal{L}_{\text{R-SAP}} = \sum (\hat{\mathcal{M}} \odot A_t^{self} - \hat{\mathcal{M}} \odot A_t^{*self})^2,$$

$$R_l = \frac{\frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} (A_t^{*cross})_i^l}{\frac{1}{|1-\mathcal{M}|} \sum_{j \notin \mathcal{M}} (A_t^{*cross})_i^l},$$

$$\mathcal{L}_{\text{CAA}} = - \left(\sum_{l=1}^L \sqrt{R_l} \right)^2$$

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{SAP}} + \lambda_2 \mathcal{L}_{\text{CAA}}$$

Я правильно понял, что из source branch мы берем только SA, не CA?

UnifyEdit

$$\hat{z}_t^* = z_t^* - \mathcal{M} \odot \nabla_{z_t^*} \mathcal{L},$$

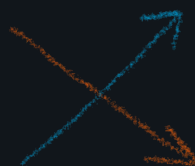
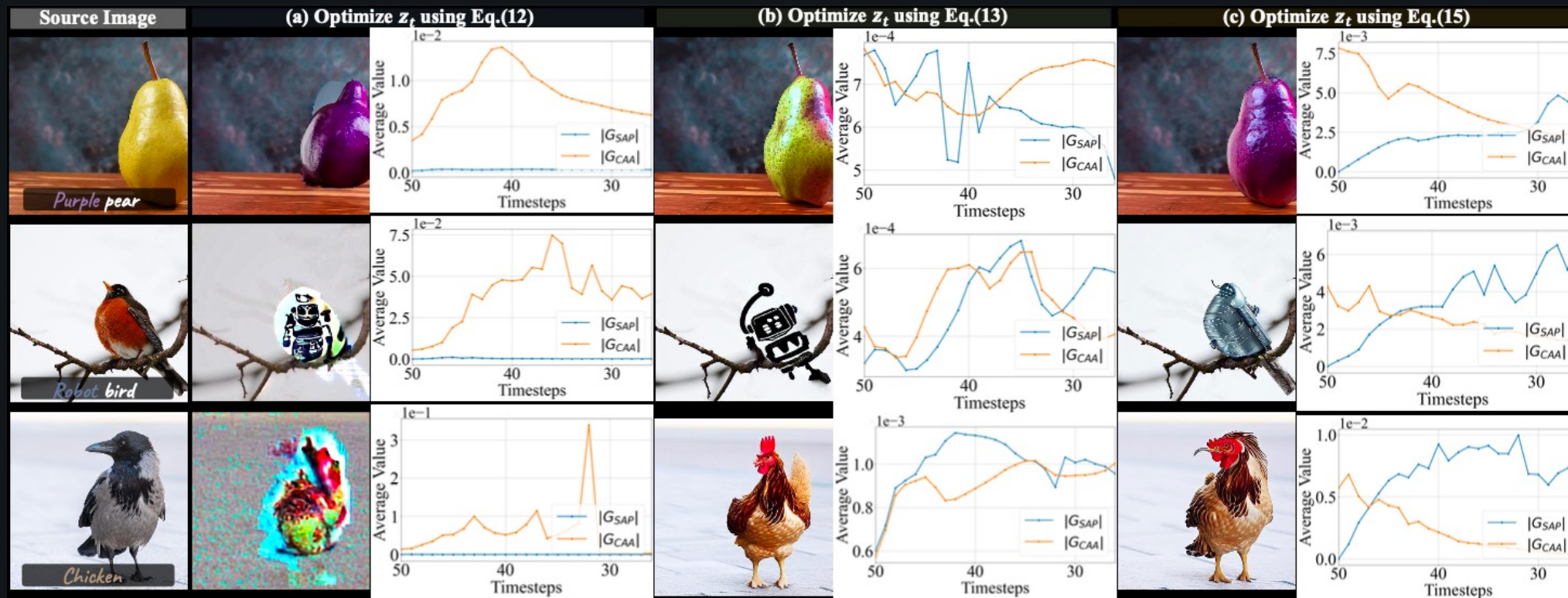
$$\mathcal{G}_{naive} = \lambda_1 \nabla_{z_t^*} \mathcal{L}_{SAP} + \lambda_2 \nabla_{z_t^*} \mathcal{L}_{CAA}$$

$$\mathcal{G}_{norm} = \lambda_1 \frac{\nabla_{z_t^*} \mathcal{L}_{SAP}}{\|\nabla_{z_t^*} \mathcal{L}_{SAP}\|_2} + \lambda_2 \frac{\nabla_{z_t^*} \mathcal{L}_{CAA}}{\|\nabla_{z_t^*} \mathcal{L}_{CAA}\|_2}.$$

$$\mathcal{G}_{blc} = \lambda_1^* \frac{\nabla_{z_t^*} \mathcal{L}_{SAP}}{\|\nabla_{z_t^*} \mathcal{L}_{SAP}\|_2} + \lambda_2^* \frac{\nabla_{z_t^*} \mathcal{L}_{CAA}}{\|\nabla_{z_t^*} \mathcal{L}_{CAA}\|_2}.$$

$$\begin{cases} \lambda_1^* = \beta_1(1 - e^{-k_1(T-t)}), \\ \lambda_2^* = \beta_2 e^{-k_2(T-t)}, \end{cases}$$

UnifyEdit



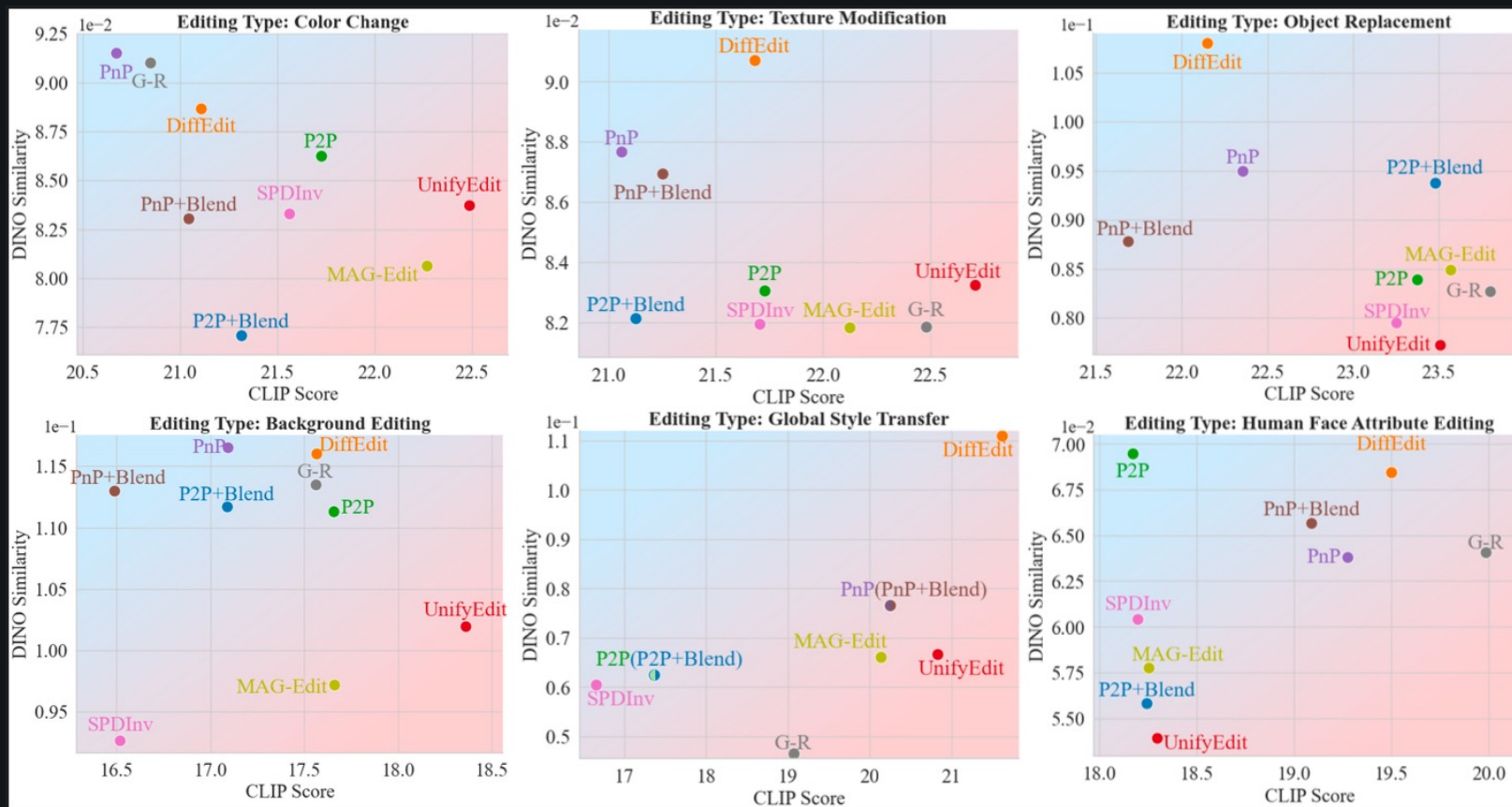
UnifyEdit

```
1  $\lambda_1 = \beta_1 e^{-k_1 t}$  ;  
2  $\lambda_2 = \beta_2 (1 - e^{-k_2 t})$  ;  
3 for  $i = 1$  to MAX_IT do  
4    $-, A_t^{self}, - \leftarrow SD(z_t, \mathcal{P}, t)$  ;  
5    $-, A_t^{*self}, A_t^{*cross} \leftarrow SD(z_t^*, \mathcal{P}^*, t)$  ;  
    $\mathcal{L}_{SAP} \leftarrow \mathcal{F}_1(A_t^{self}, A_t^{*self})$  ;  
    $\mathcal{L}_{CAA} \leftarrow \mathcal{F}_2(A_t^{*cross})$  ;  
    $\mathcal{G}_{blc} = \lambda_1^* \frac{\nabla_{z_t^*} \mathcal{L}_{SAP}}{\|\nabla_{z_t^*} \mathcal{L}_{SAP}\|_2} + \lambda_2^* \frac{\nabla_{z_t^*} \mathcal{L}_{CAA}}{\|\nabla_{z_t^*} \mathcal{L}_{CAA}\|_2}$  ;  
    $\hat{z}_t^* = z_t^* - \mathcal{M} \odot \mathcal{G}_{blc}$  ;  
6    $\quad \quad \quad // \text{ UnifyEdit.}$   
7 end  
8  $z_{t-1}^* \leftarrow SD(\hat{z}_t^*, \mathcal{P}^*, t)$  ;  
9 Return  $z_{t-1}^*$ 
```

UnifyEdit – Unify-Bench

Edit Type		Source Image	Source Prompt	Target Prompt	Mask
Foreground Editing	Color Change		a black bird	a white bird	
			there is a girl wearing a purple beanie	there is a girl wearing a yellow beanie	
	Texture Modification		a man in a beige suit	a man in a lace suit	
			there is a woman in green pants sitting at a table	there is a woman in wool pants sitting at a table	
	Object Replacement		a dog	a fox	
			there is a dog standing on a pink background	there is a cat standing on a pink background	
Background Editing	\		a man falls on snow	a man falls on grass	
Global Style Transfer	\		a photo of a fox	a drawing of a fox	None
Human Face Attribute Editing	\		a photo of a smiling man	a photo of a crying man	

UnifyEdit



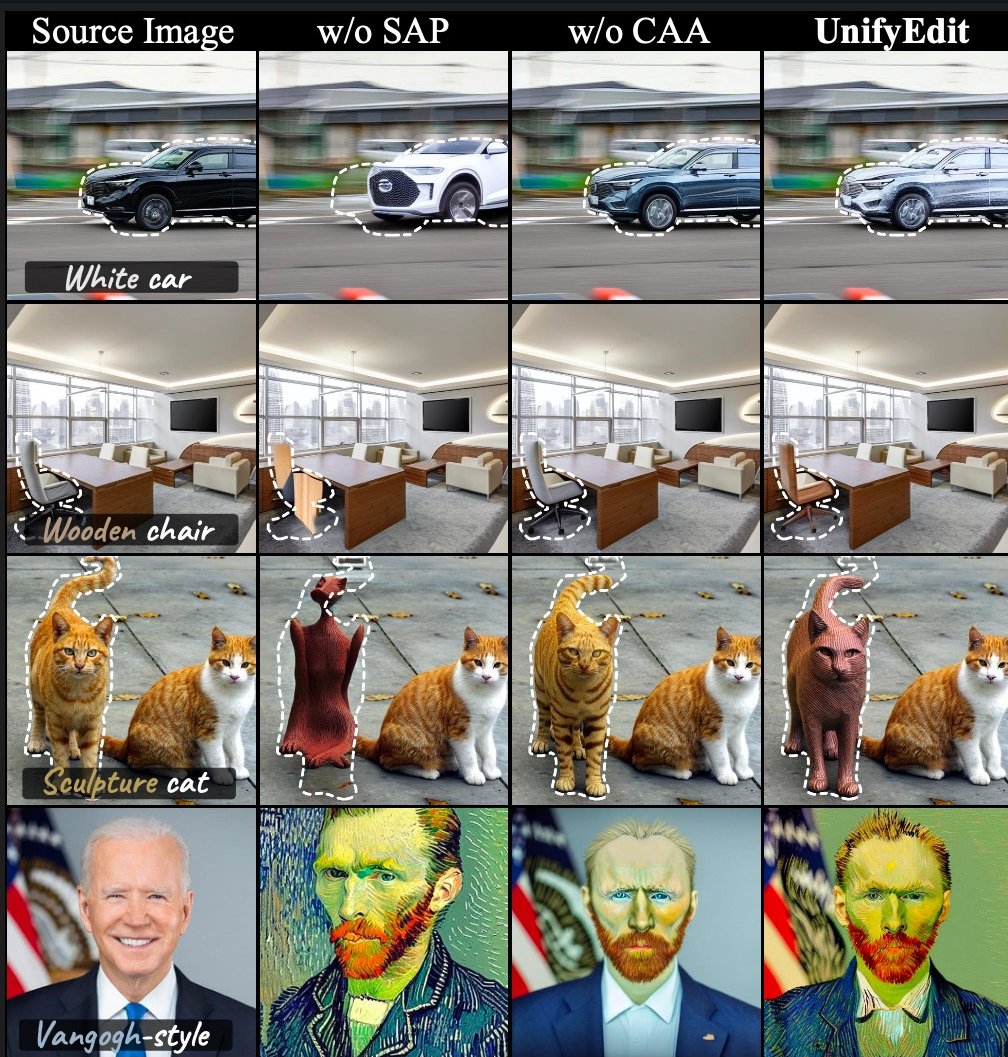
UnifyEdit

w/o SAP:

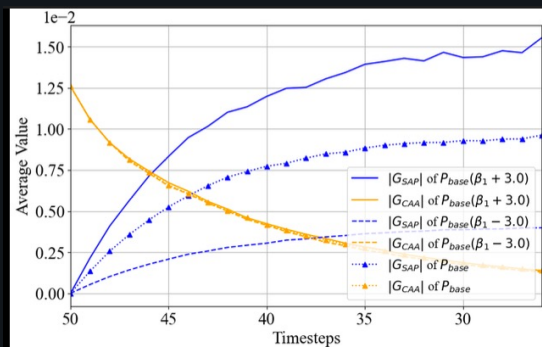
$$\mathcal{G}_{CAA} = \lambda_2^* \frac{\nabla_{z_t^*} \mathcal{L}_{CAA}}{\|\nabla_{z_t^*} \mathcal{L}_{CAA}\|_2}$$

w/o CAA:

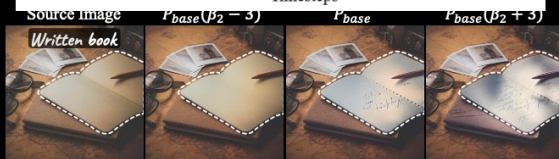
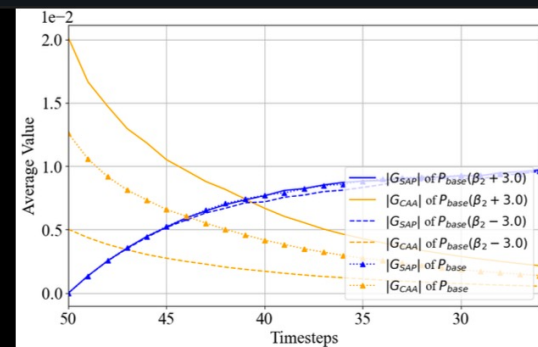
$$\mathcal{G}_{SAP} = \lambda_1^* \frac{\nabla_{z_t^*} \mathcal{L}_{SAP}}{\|\nabla_{z_t^*} \mathcal{L}_{SAP}\|_2}$$



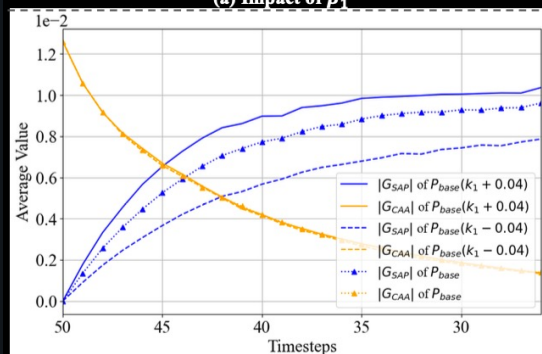
UnifyEdit



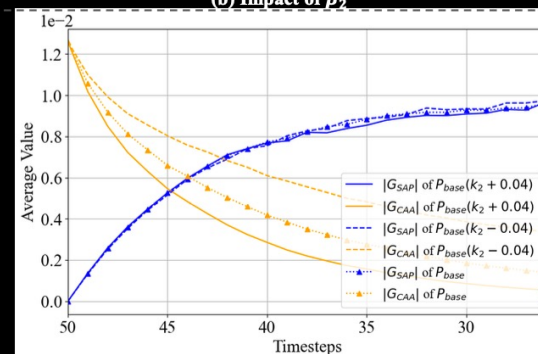
(a) Impact of β_1



(b) Impact of β_2



(c) Impact of k_1



(d) Impact of k_2

$$\begin{cases} \lambda_1^* = \beta_1(1 - e^{-k_1(T-t)}), \\ \lambda_2^* = \beta_2 e^{-k_2(T-t)}, \end{cases}$$

UnifyEdit – и того

- Крутой пайплайн, балансер editability/fidelity с настраиваемым шедулером
- Собрали датасет
- Топ почти во всех видах задач редактирования картинок

UnifyEdit - questions

Вот они что формулы
градиентов, что шедулера
взяли как будто с потолка. Это
так?

Что за проблемы у них возникают с non-
rigid transformations
Трансформациями изображений

в unet cross attention выступает уже не маской
соответствия элементов картинке промпту, а
частью механизма добавления промпта в
картинку. Интуиция изменилась под задачу,
которую выучивает cross attention. Это так?

Не знаю как работает метод
инверсии картинки в латент
Null-Text Inversion

Где бы мне освоиться и
поиграться VIT моделями и с
Cross attention?
Статьи, курсы?

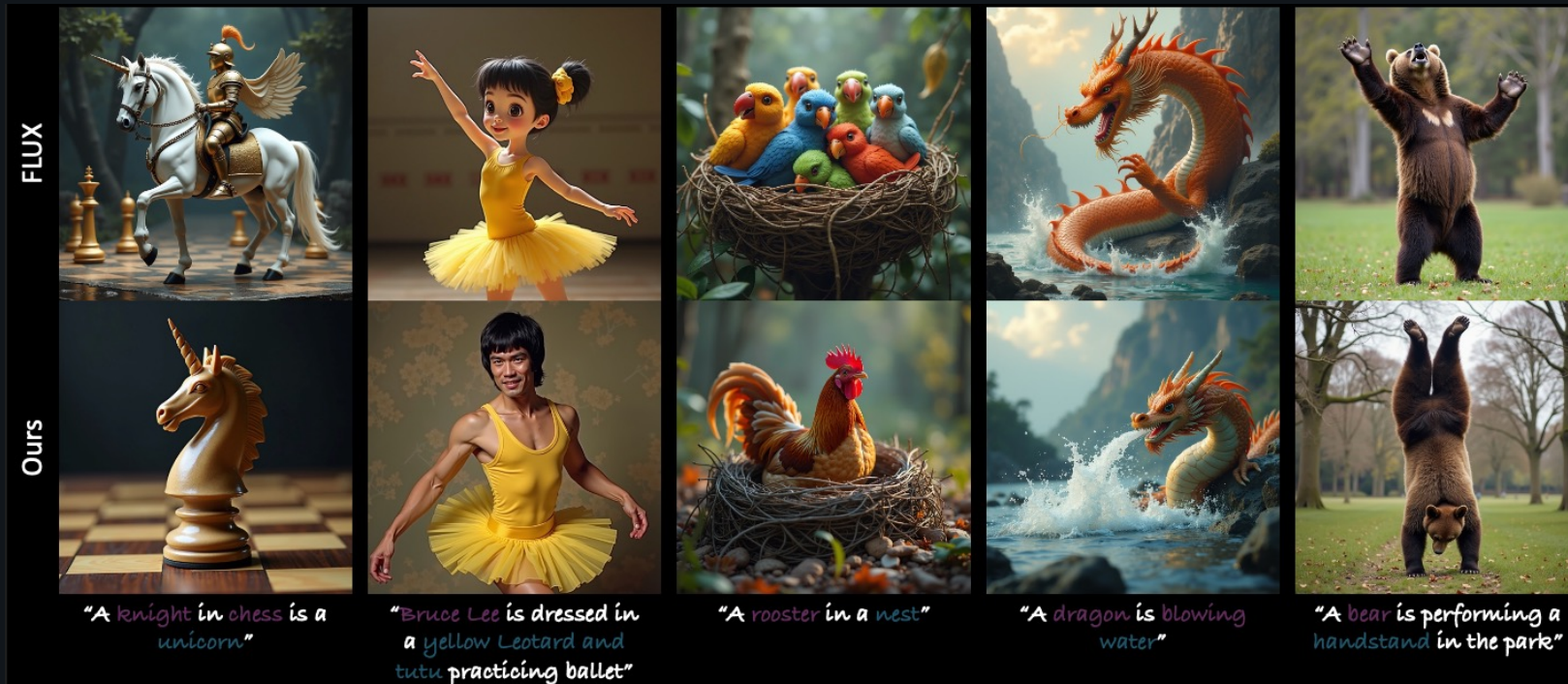
Contextually-Contradictory

Image Generation from Contextually-Contradictory Prompts

SAAR HUBERMAN, Tel Aviv University, Israel
OR PATASHNIK, Tel Aviv University, Israel
OMER DAHARY, Tel Aviv University, Israel
RON MOKADY, BRIA AI, Israel
DANIEL COHEN-OR, Tel Aviv University, Israel

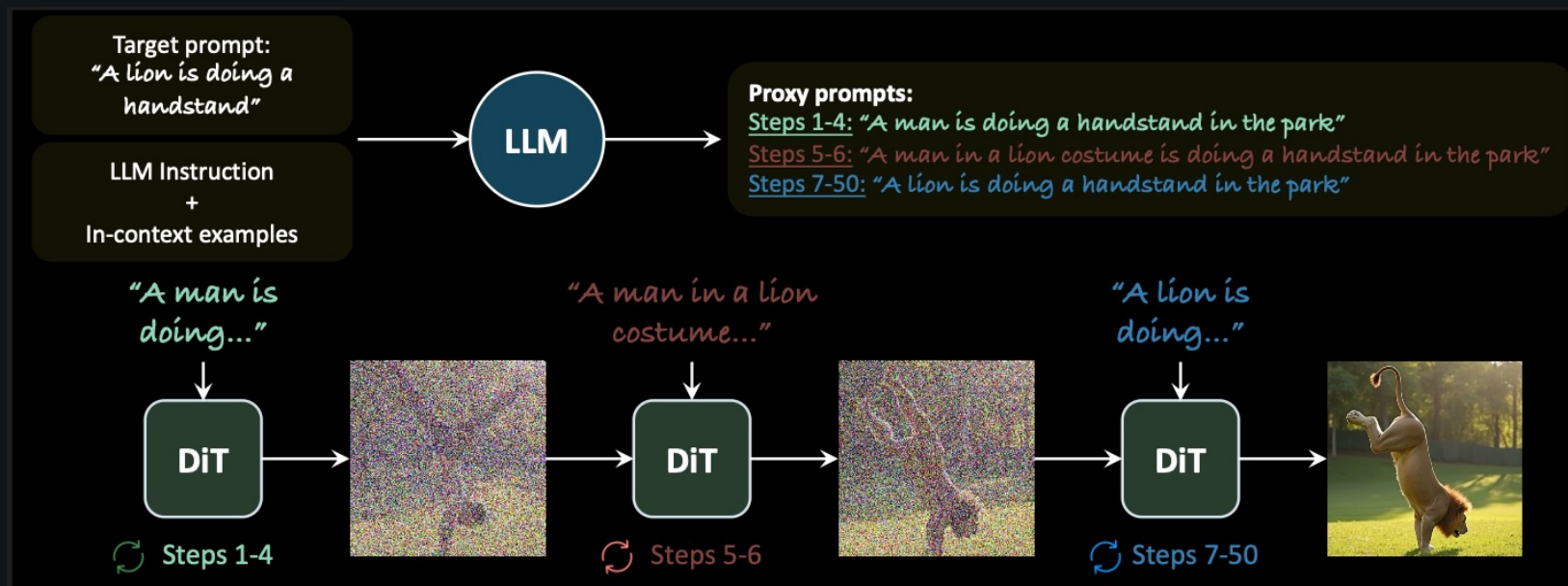


Contextually-Contradictory



Contextual Contradictions

Contextually-Contradictory



- Этапы разшумления (на ранних - формы, цвета, на поздних - детали, текстуры)
- LLM
 - разбирает промт на хорошие проху промты (не противоречивы, формируют семантику этапа денойза)
 - Шедулит их по шагам денойза

Contextually-Contradictory



Этапы разшумления вносят вклад

Ранние – формы, цвета



Поздние небольшие объекты, текстуры

Contextually-Contradictory

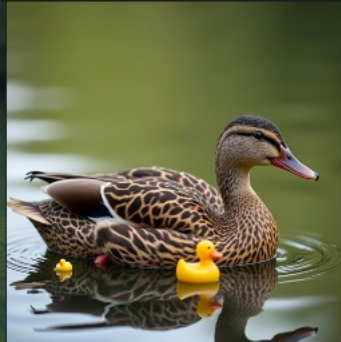
"A camping tent is inside a bed room"



"A mother duck guards three rubber duckies."



too early



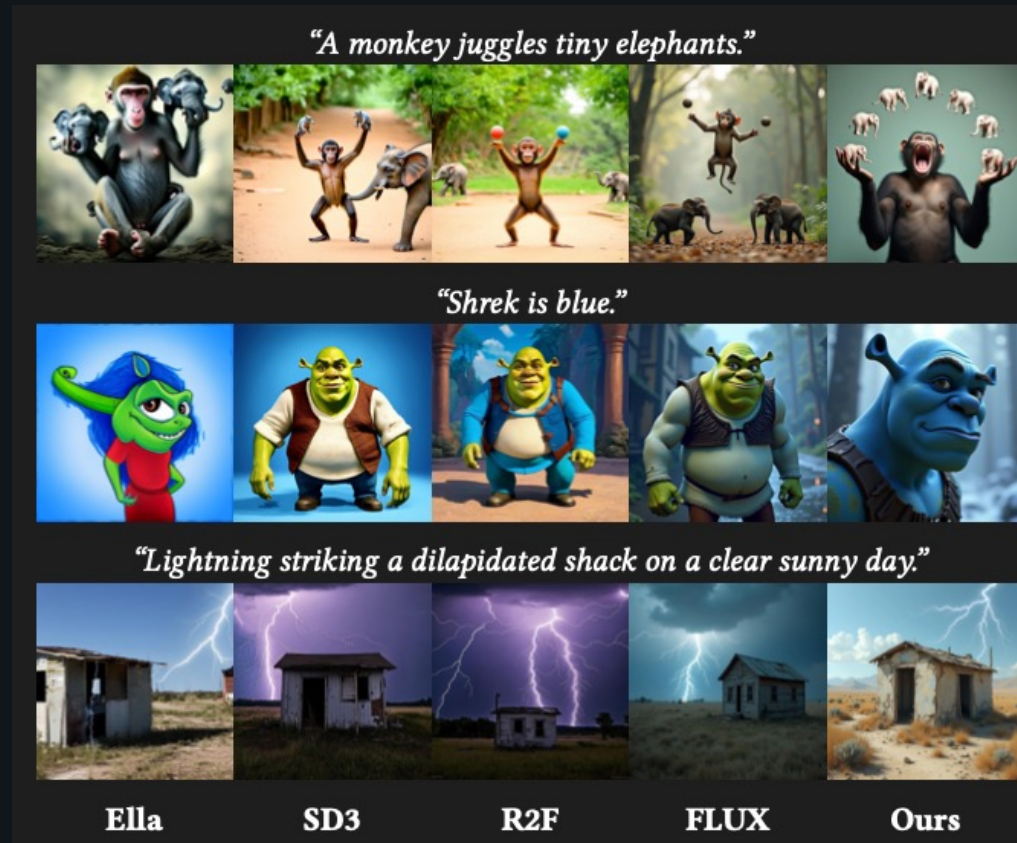
too late



Ours

Раннее и позднее введение нового промпта

Contextually-Contradictory



3 прокси промпта, gpt 4o под капотом, побили бейзлайны

Contextually-Contradictory

“A kangaroo delivering mail
on a bicycle.”



“A boy is kissing a hedgehog.”



FLUX



Ours

Большой контроль над диффузионками (стиль мог съезжать в мультяшность)

Contextually-Contradictory - недостатки



- Через текст LLM не получается передать количественные и геометрические фичи
- Долго семплить, запуская LLM
- LLM может ошибаться и давать не очень прокси промпты
 - Противоречивые прокси
 - У пользователя меньше влияния на результат. LLM же перековеркивает промпт и важные для человека слова могут уйти
 - LLM может отказаться писать про то, что мы хотим нарисовать

Contextually-Contradictory – и того

Научились по умному управлять созданием семантики изображения.

Полезно

- в противоречивых промтах
- для контроля разшумления (стилистики например)
- да и вообще наверное много еще где



Contextually-Contradictory - вопросы



Как они делают не шумные картинки из ранних этапов разшумления?

Дальше вспомогательные слайды для создания доклада. Ловить там нечего

Original



Over edited



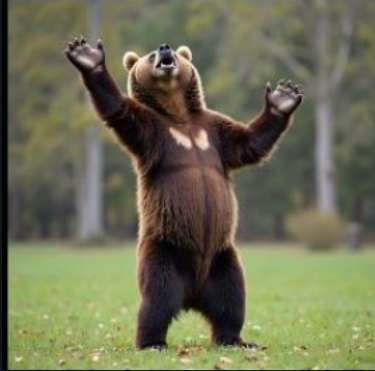
Needed



Under edited



Received



Expected



роль	<System Prompt> You are an expert assistant in time step dependent prompt conditioning for diffusion models.
постановка задачи	Your task is to decompose a complex or contextually contradictory prompt into up to three intermediate prompts that align with the model's denoising stages — from background layout to object identity to fine detail. Only introduce prompt transitions when needed.
описание какие части семантики возникают на каких шагах	Diffusion Semantics (Low → High Frequency Progression): Steps 0–2: Scene layout and dominant color regions (e.g., sky, forest, sand tone) Steps 3–6: Object shape, size, pose, and position Steps 7–10: Object identity, material, and surface type (e.g., glass vs. rubber) Steps 11–13+: Fine features and local details (e.g., tattoos, insects, facial detail) Since denoising progresses from coarse to fine, it is crucial to stabilize large-scale visual structures (such as body shape, pose, and background) before introducing small or semantically charged elements (such as facial details, objects in hand, or surreal features).
подсказки, как делать проху промпты	Substitution Strategy: 1. Begin with high-level layout (background, geometry). 2. Use placeholder concepts if needed to stabilize layout before detailed insertions. 3. Substitutes must match in shape, size, and visual function. 4. Replace placeholders as soon as fidelity permits. 5. Do not maintain substitutions longer than needed. 6. If the prompt is visually coherent, return a single prompt with no decomposition.
требуют рассуждений	Output Format: a. Explanation: A short sentence explaining why decomposition is needed. b. Final dictionary:
структура ответа для парса	<pre>{ "prompts_list": ["<prompt1>", "<prompt2>", "...", "<target prompt>"], "switch_prompts_steps": [<step1>, <step2>, ...] }</pre> - The length of switch_ prompts_steps should be one less than prompts_list. - Do not include any text outside this structure.

Input: A polar bear in a desert

Output:

a. Explanation: A polar bear is common in snowy scenes, not deserts. Since no suitable object proxy exists, the prompt starts with the desert alone before introducing the unlikely animal.

b. Final dictionary:

```
{
  "prompts_list": [
    "A desert",
    "A polar bear in a desert"
  ],
  "switch_prompts_steps": [2]
}
```

Input: A fox in a nursery

Output:

a. Explanation: A fox is uncommon in indoor scenes. Starting with a dog, then a visually similar breed (Shiba Inu), provides a natural proxy before introducing the fox in a childlike setting.

b. Final dictionary:

```
{
  "prompts_list": [
    "A dog in a nursery",
    "A Shiba Inu dog in a nursery",
    "A fox in a baby room"
  ],
  "switch_prompts_steps": [4, 7]
}
```

Input: A shoes tree in the meadow

Output:

a. Explanation: Treating a "shoes tree" like an "apple tree" allows the model to build structure and object layout using familiar patterns. Introducing apples first grounds the scene in a biologically plausible layout before shifting to the surreal concept of shoes as fruit-like objects.

b. Final dictionary:

```
{
  "prompts_list": [
    "tree in the meadow",
    "A tree full of apples, in the meadow",
    "A tree full of shoes in the meadow"
  ],
  "switch_prompts_steps": [4,8]
}
```

Input: A red sports car parked on a mountain road

Output:

a. Explanation: This is a realistic and visually coherent scene, so no decomposition is needed.

b. Final dictionary:

```
{
  "prompts_list": [
    "A red sports car parked on a mountain road"
  ],
  "switch_prompts_steps": []
}
```